

314707003_11.24.R

CDH

2026-05-11

```
rm(list=ls())  
#install.packages("AER")  
library(P0E5Rdata)
```

```
##  
## 載入套件：'P0E5Rdata'
```

```
## 下列物件被遮斷自 'package:datasets':  
##  
##      euro
```

```
library(AER)
```

```
## Warning: 套件 'AER' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：car
```

```
## Warning: 套件 'car' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：carData
```

```
## Warning: 套件 'carData' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：lmtest
```

```
## Warning: 套件 'lmtest' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：zoo
```

```
## Warning: 套件 'zoo' 是用 R 版本 4.5.3 來建造的
```

```
##  
## 載入套件：'zoo'
```

```
## 下列物件被遮斷自 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
## 載入需要的套件: sandwich
```

```
## Warning: 套件 'sandwich' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件: survival
```

```
library(car)  
data = fultonfish  
  
#11.24(a)  
rf_price <- lm(lprice ~ mon + tue + wed + thu + stormy + mixed, data = fultonfish)  
# check p-value of mixed  
coef(summary(rf_price))
```

```
##              Estimate Std. Error    t value    Pr(>|t|)  
## (Intercept) -0.35956847 0.07915415 -4.5426355 1.501702e-05  
## mon         -0.10824827 0.10338898 -1.0470000 2.975268e-01  
## tue         -0.06609207 0.10103460 -0.6541528 5.144573e-01  
## wed         -0.04927356 0.10377370 -0.4748174 6.359127e-01  
## thu          0.03934669 0.10070504  0.3907122 6.968092e-01  
## stormy       0.44633866 0.07921108  5.6348008 1.508139e-07  
## mixed        0.23687528 0.07851343  3.0170036 3.209729e-03
```

```
# check the joint significance of STORMY and MIXED  
linearHypothesis(rf_price, c("stormy = 0", "mixed = 0"))
```

```
##  
## Linear hypothesis test:  
## stormy = 0  
## mixed = 0  
##  
## Model 1: restricted model  
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed  
##  
##    Res.Df    RSS Df Sum of Sq    F    Pr(>F)  
## 1      106 15.876  
## 2      104 12.115  2     3.7604 16.14 7.857e-07 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#11.24(b)  
demand_iv <- ivreg(lquan ~ lprice + mon + tue + wed + thu |  
  mon + tue + wed + thu + stormy + mixed, data = fultonfish)  
coef(summary(demand_iv))
```

```
##               Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  8.54023437  0.1566081  54.53250335  7.415812e-79
## lprice      -0.93014092  0.3533995  -2.63198161  9.768165e-03
## mon         -0.01190203  0.2083710  -0.05711941  9.545586e-01
## tue         -0.52583152  0.2022954  -2.59932461  1.068445e-02
## wed         -0.56261965  0.2069581  -2.71851907  7.673542e-03
## thu          0.09987059  0.2028454   0.49234827  6.235011e-01
## attr(,"df")
## [1] 105
## attr(,"nobs")
## [1] 111
```

```
#11.24(c)
u_hat <- resid(demand_iv)
sargan_aux <- lm(u_hat ~ mon + tue + wed + thu + stormy + mixed, data = fulto
nfish)
S <- nobs(demand_iv) * summary(sargan_aux)$r.squared
p_value <- pchisq(S, df = 1, lower.tail = FALSE)
round(c(Sargan = S, df = 1, p.value = p_value), 4)
```

```
## Sargan      df p.value
## 0.7722  1.0000  0.3795
```

```
#11.24(d)
linearHypothesis(rf_price, c("mon = 0", "tue = 0", "wed = 0", "thu = 0"))
```

```
##
## Linear hypothesis test:
## mon = 0
## tue = 0
## wed = 0
## thu = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     108 12.408
## 2     104 12.115   4   0.29256 0.6279 0.6437
```

314707003_11.28.R

CDH

2026-05-12

```
rm(list=ls())  
#install.packages("AER")  
library(POE5Rdata)
```

```
##  
## 載入套件：'POE5Rdata'
```

```
## 下列物件被遮斷自 'package:datasets':  
##  
##      euro
```

```
library(AER)
```

```
## Warning: 套件 'AER' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：car
```

```
## Warning: 套件 'car' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：carData
```

```
## Warning: 套件 'carData' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：lmtest
```

```
## Warning: 套件 'lmtest' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：zoo
```

```
## Warning: 套件 'zoo' 是用 R 版本 4.5.3 來建造的
```

```
##  
## 載入套件：'zoo'
```

```
## 下列物件被遮斷自 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
## 載入需要的套件: sandwich
```

```
## Warning: 套件 'sandwich' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件: survival
```

```
library(car)  
data = truffles  
#11.28(a) 略  
  
#11.28(b)  
#demand  
dem_iv <- ivreg(p ~ q + ps + di | ps + di + pf, data = truffles)  
coef(summary(dem_iv))
```

```
##              Estimate Std. Error   t value    Pr(>|t|)  
## (Intercept) -11.428407  13.591606 -0.840843 4.081026e-01  
## q            -2.670519   1.174955 -2.272869 3.153505e-02  
## ps           3.461081   1.115572  3.102517 4.582228e-03  
## di          13.389921   2.746707  4.874899 4.675236e-05  
## attr(,"df")  
## [1] 26  
## attr(,"nobs")  
## [1] 30
```

```
#supply  
sup_iv <- ivreg(p ~ q + pf | ps + di + pf, data = truffles)  
coef(summary(sup_iv))
```

```
##              Estimate Std. Error   t value    Pr(>|t|)  
## (Intercept) -58.798223  5.8591611 -10.03526 1.316492e-10  
## q            2.936711   0.2157717  13.61027 1.320944e-13  
## pf           2.958486   0.1559638  18.96905 3.879687e-17  
## attr(,"df")  
## [1] 27  
## attr(,"nobs")  
## [1] 30
```

```
#11.28(c)  
Pbar <- mean(truffles$p)  
Qbar <- mean(truffles$q)  
elas <- (1 / coef(dem_iv)["q"]) * (Pbar / Qbar)  
round(c(Pbar = Pbar, Qbar = Qbar, elasticity = elas), 4)
```

```
##          Pbar          Qbar elasticity.q
##        62.7240        18.4583        -1.2725
```

```
#11.28(d)
PSF <- 22
DIF <- 3.5
PFF <- 23

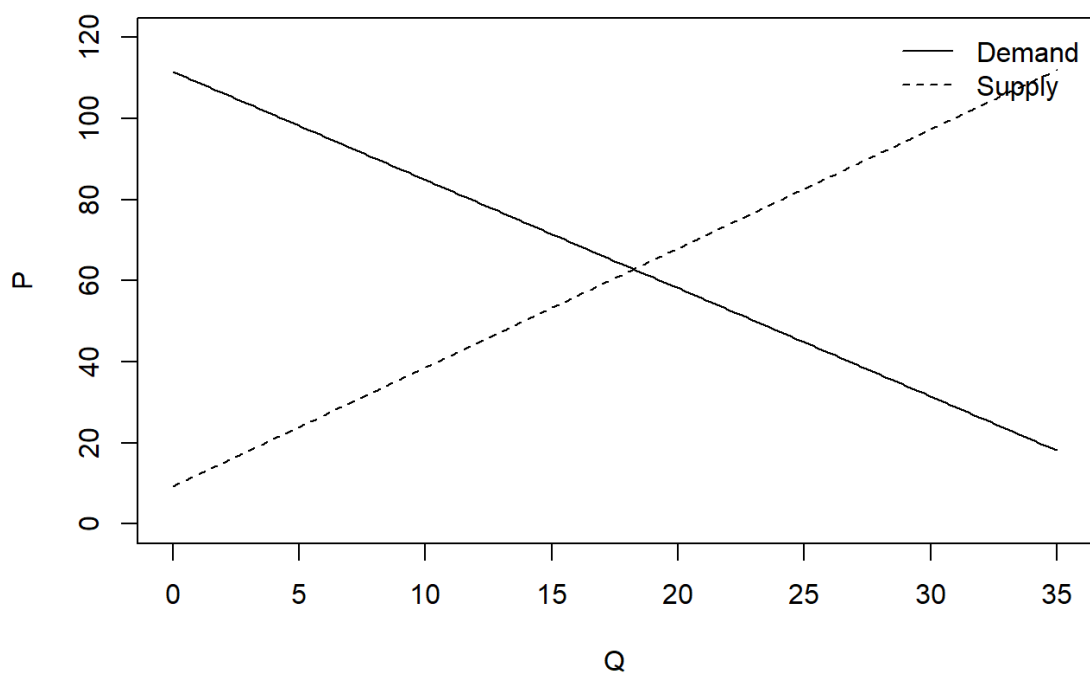
d_int <- coef(dem_iv)[1] + coef(dem_iv)["ps"] * PSF + coef(dem_iv)["di"] * DI
F
s_int <- coef(sup_iv)[1] + coef(sup_iv)["pf"] * PFF

round(c(
  D_intercept = unname(d_int),
  D_slope_Q   = unname(coef(dem_iv)["q"]),
  S_intercept = unname(s_int),
  S_slope_Q   = unname(coef(sup_iv)["q"])
), 4)
```

```
## D_intercept  D_slope_Q S_intercept  S_slope_Q
##    111.5801    -2.6705     9.2470     2.9367
```

```
#畫圖
Qgrid <- seq(0, 35, length.out = 200)
plot(
  Qgrid, d_int + coef(dem_iv)["q"] * Qgrid,
  type = "l",
  xlab = "Q",
  ylab = "P",
  ylim = c(0, 120),
  main = "Truffle Demand and Supply"
)
lines(
  Qgrid, s_int + coef(sup_iv)["q"] * Qgrid,
  lty = 2
)
legend(
  "topright",
  legend = c("Demand", "Supply"),
  lty = c(1, 2),
  bty = "n"
)
```

Truffle Demand and Supply



```
#11.28(e)
Q_eq <- (d_int - s_int) / (coef(sup_iv)["q"] - coef(dem_iv)["q"])
P_eq <- d_int + coef(dem_iv)["q"] * Q_eq
round(c(P_eq = unname(P_eq), Q_eq = unname(Q_eq)), 4)
```

```
##      P_eq      Q_eq
## 62.8426 18.2502
```

```
rf_q <- lm(q ~ ps + di + pf, data = truffles)
rf_p <- lm(p ~ ps + di + pf, data = truffles)
round(coef(summary(rf_q)), 4)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.8951      3.2434   2.4342  0.0221
## ps             0.6564      0.1425   4.6051  0.0001
## di             2.1672      0.7005   3.0938  0.0047
## pf            -0.5070      0.1213  -4.1809  0.0003
```

```
round(coef(summary(rf_p)), 4)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -32.5124      7.9842  -4.0721  4e-04
## ps           1.7081      0.3509   4.8682  0e+00
```

```
## di          7.6025      1.7243  4.4089      2e-04
## pf          1.3539      0.2985  4.5356      1e-04
```

```
newdata <- data.frame(ps = 22, di = 3.5, pf = 23)
Q_rf <- predict(rf_q, newdata = newdata)
P_rf <- predict(rf_p, newdata = newdata)
round(c(Q_rf = unname(Q_rf), P_rf = unname(P_rf)), 4)
```

```
##      Q_rf      P_rf
## 18.2604 62.8154
```

```
#compare
round(rbind(
  structural_2SLS = c(Q = unname(Q_eq), P = unname(P_eq)),
  reduced_form    = c(Q = unname(Q_rf), P = unname(P_rf)),
  difference       = c(Q = unname(Q_eq - Q_rf), P = unname(P_eq - P_rf))
), 4)
```

```
##              Q      P
## structural_2SLS 18.2502 62.8426
## reduced_form    18.2604 62.8154
## difference      -0.0102  0.0272
```

```
#11.28(f)
sup_ols <- lm(p ~ q + pf, data = truffles)
round(coef(summary(sup_ols)), 4)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -52.8763      5.0238 -10.5252      0
## q            2.6613      0.1712  15.5436      0
## pf           2.9217      0.1482  19.7150      0
```

```
dem_ols <- lm(p ~ q + ps + di, data = truffles)
round(coef(summary(dem_ols)), 4)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -13.6195      9.0872 -1.4987    0.1460
## q            0.1512      0.4988  0.3032    0.7642
## ps           1.3607      0.5940  2.2909    0.0303
## di           12.3582      1.8254  6.7701    0.0000
```